The Aging Challenge: A Performance Evaluation on Baseline Face Recognition Models Under Cross Age Variation

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***Abstract*—Facial recognition systems have gained significant prominence over the past few years due to their enhanced security features and operational efficiency. These systems offer a secure and transparent method for identity verification, reducing reliance on physical documents which are often prone to theft. However, these face recognition systems struggle to identify the same individual with large age gaps. This is because face recognition systems try to capture unique facial features like distance between eyes, skin structure, bone structure etc. which can greatly be affected due to aging which the models are often not trained to handle. This paper aims to deliver a comprehensive analysis of how different baseline face recognition models perform with respect to identification and verification tasks when tested with age variant datasets like Cross-Age Celebrity Dataset (CACD) and FG-NET. We benchmarked four baseline face recognition models - ArcFace, OpenFace, FaceNet, VGGFace, evaluating their performance using rank metrics for identification tasks and AUC, EER metrics for verification tasks. Our key contributions include : (1) a comprehensive benchmarking of four baseline face recognition models on age variant datasets using identification and verification protocols; (2) identification of specific age groups where performance degrades; and (3) empirical evidence indicating that VGGFace demonstrates stronger performance under cross age face recognition scenarios.**

***Index Terms*—Cross Age Verification, Face Recognition, Identification, Verification**

I. Introduction

Facial recognition systems have become widely adopted for identification and verification tasks, in several domains such as airport security, unlocking personal devices, authorization of transactions, etc.There are multiple factors that contribute to the rise of facial recognition systems as a medium for verification. Facial features are unique to each individual and difficult to replicate, which makes them a perfect fit to verify a person's identity. Additionally, these systems are highly convenient for users to authenticate themselves quickly without the need for any physical credentials.

However, despite their widespread use, facial recognition systems face significant challenges in recognizing an individual's face as it changes over time due to aging [1]. This limitation poses a significant hurdle in law enforcement, where a suspect's faces are matched with existing crime records to identify the criminal. It can also cause hindrance in airport security systems as passports in some countries are valid for about a decade. In such cases the age induced facial changes can lead to false rejections. Thereby slowing down passenger flow and creating an inefficient system [2].

Furthermore, face recognition models are usually trained on datasets with limited age diversity leading to their poor performance in handling age variations. This study aims to benchmark baseline face recognition models using age variant datasets and appropriate evaluation protocols. Through this, we intend to highlight the limitations of existing systems, which will contribute to creation of more robust and age invariant face recognition solutions.

II. Literature Review

Several studies have examined how facial aging impacts recognition accuracy of face recognition models. This section reviews relevant research addressing the problem of age-related variations in face recognition.

Studies have addressed the challenge of age variation in face recognition by developing models that separate identity and age-related features. [3] introduced orthogonal embeddings CNNS which decompose facial features into two orthogonal components - in which one represents age related features and other identity related features, of which the identity related features can be used for the purpose of face recognition. [4] proposes an approach that will remove age related features through adversarial factorization reducing mismatch. More recent work [5], such as MTLFace, combines face age synthesis with age related representations, to yield better results rather than treating them separately.

Recent advancements in face recognition technology incorporate transformer-based and hybrid architecture [6],[7],[8], which achieves superior performance by modeling long range dependencies. This study, however, focuses on evaluating performance of widely adopted CNN based face recognition approaches- OpenFace, FaceNet, VGGFace and ArcFace. The selection of these models is motivated by several factors. These

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models are foundational architectures deployed in various real-world applications and serve as a benchmark for industry standards. Despite their prevalence, there has been limited literature about the performance of these models. Establishing performance baselines with well understood architectures will help us establish benchmarks that guide evaluation of newer architecture, including transformer-based approaches in future work.

III. Methodology

This paper aims to evaluate the performance of four state-of-the-art face recognition models - ArcFace, OpenFace, FaceNet, VGGFace on age diverse data to identify the specific age groups and scenarios where performance declines. These four models were chosen as they represent diverse architectural designs and training approaches allowing for a fair and balanced comparison. For this purpose, two publicly available datasets CACD and FG-NET were selected due to their diverse nature and inclusion of images taken in unconstrained environments. The model's performance was tested against both identification and verification tasks to assess model's robustness against age related variations.

A. Data Description

In this study we utilize two widely used datasets, CACD and FG-NET for evaluation. The CACD dataset contains 163,446 images of 2,000 subjects, a large volume suitable for a more robust evaluation. The FG-NET dataset contains 1002 images of 82 subjects from age groups 0-69 years.The rationale for selecting these datasets is that they allow us to rigorously evaluate the performance of state-of-the-art face recognition models across real world conditions.

B. Task Definitions

Face recognition models are typically evaluated on two fundamental tasks- identification and verification. Identification refers to a 1:N problem, where the model attempts to identify an individual by matching their face against a known database of faces to find a potential match. This task is commonly applied in surveillance and law enforcement domains. Verification on the other hand is a 1:1 problem, in which we try to confirm if a face matches a specific face on the record. This is primarily used in applications such as biometric authentication and online identity verification.

1) Identification Task Protocol: The two common protocols used for identification tasks - closed set identification and open set identification. In a closed set protocol, we assume that every face to be identified is present in the system's database. The model's task is to find the best match from the known identities of the database. This is useful for testing in controlled environments where the subjects are limited and fixed. On the other hand, open set protocol, we assume that the face to be identified may or may not be present in our database. If the facial input is present in the database , the model retrieves their identity, if face is not present it simply flags them as unknown.

For this study we adopt a closed set protocol, as it provides a controlled evaluation setup. Allowing for a focused investigation of model's performance related to age variation rather than the additional challenge of rejecting unknown identities.

The primary evaluation metrics for this task would be rank based metrics. Rank-k accuracy measures if the identity of the provided face is within the top k predicted matches given by the model.For each identity a similarity score is computed , and we sort the images in dataset by this score and check if the true identity is among top k matches. Rank-k accuracy is defined as:

$$\text{Rank-}k = \frac{\text{Number of correct matches within top-}k}{N} \tag{1}$$

where N denotes number of probe images

2) Verification Task Protocol: Verification evaluation commonly uses three protocols- all versus all, k -fold cross validation and subject-disjoint. In all versus all protocol, every image in the test set is compared to every image in the training set and a similarity score is generated for each pair. These pairs are then classified into either match (same person) or not match (different person). This protocol is computationally expensive and does not ensure fair generalization.

Cross validation partitions the dataset into k folds. The model is then trained on k-1 folds and then tested on the remaining one-fold. This process is repeated k times so that each set acts as a test set once [9, Sec. 7.10]. However this approach poses a significant risk of data leakage as the method randomly splits the dataset into k subsets, the same image could appear in both training and test sets thereby inflating performance.

To address this limitation, we employ the subject disjoint protocol which ensures that training and testing sets are disjoint. It does so by first grouping the entire dataset by identity and then splitting these groups of people into training or testing splits. This ensures a fair evaluation of generalization under age variation and is also consistent with standard benchmarking practices [10],[11],[12],[13].

Performance is measured for this task using AUC which is the area under the ROC curve, where ROC plots True Positive Rate (TPR) vs False Positive Rate (FPR). Indicating the model's ability to differentiate between genuine (faces of the same person) and impostor (faces of different people) pairs. A higher value of AUC indicates better performance.

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR}) \tag{2}$$

Equal error rate (EER) refers to the threshold where false acceptance rate (FAR) and false rejection rate (FRR) become equal. EER is an important metric in biometric systems as it measures the efficiency of the system in identifying individuals correctly [11]. The lower the EER the better the system.

$$\text{EER} = \text{FAR}(\tau^*) = \text{FRR}(\tau^*) \tag{3}$$

where τ^* denotes the decision threshold at which FAR equals FRR.

C. Data Preparation and Embedding Extraction

To ensure consistent evaluation across different models, all images were preprocessed according to each model's requirements. (eg, 160x160 for FaceNet, 224x224 for VGGFace, etc.). Fixed length embeddings were extracted from each pre-trained model. In total 1002 embeddings by each model were extracted for FG-NET. From the original CACD dataset containing 163,446 images, embeddings were extracted for 152,948. This discrepancy arises primarily due to few corrupted or unreadable images within the dataset. Such cases were skipped to maintain data quality.

D. Performance Evaluation Setup

To evaluate the model on identification tasks we adopted a closed set identification protocol. Embeddings extracted from each model were compared using cosine similarity. The embedding set was divided into gallery and probe samples, and each probe embedding was matched against the gallery and the identities were ranked by similarity. Rank k accuracy metrics (with k values 1, 5 and 10) were computed by ranking gallery matched for each probe checking if the true identity appeared within top k retrieved results.

For verification, we implemented a subject- disjoint verification protocol. The dataset was first processed to create discrete age groups (bins) representing different age groups (e.g. 0-11, 12-17, 18-29, etc.). This grouping allows us to assess the model's performance within and across age groups. Positive (images from same subject) and negative pairs (images of different subjects) were generated, and cosine similarity was computed between pairs. Evaluation was done based on standard metrics like Area under ROC Curve (AUC) and Equal Error rate (EER) for each age-bin pairing. This methodology enables us to gain detailed insights about performance across various age gaps and ensures rigorous validation by avoiding data leakage.

IV. Results

This section discusses the results obtained for identification and verification tasks across four baseline models- ArcFace, FaceNet, VGGFace and OpenFace on CACD and FG-NET datasets.

A. Identification Performance

Identification accuracy was evaluated for four baseline models- ArcFace, FaceNet, VGGFace and OpenFace on FG-NET and CACD datasets at Rank-1, Rank-5, and Rank-10. The results are summarized in Table 1 and 2 , and visualized in Figure 1 and 2, providing a comparative overview of each model's performance at different ranks and datasets.

Rank-1 accuracy is extremely low for both the datasets , especially FG-NET due to its large age progression. A detailed explanation for this poor performance and its relation to transitions from childhood to adulthood is given in section V.

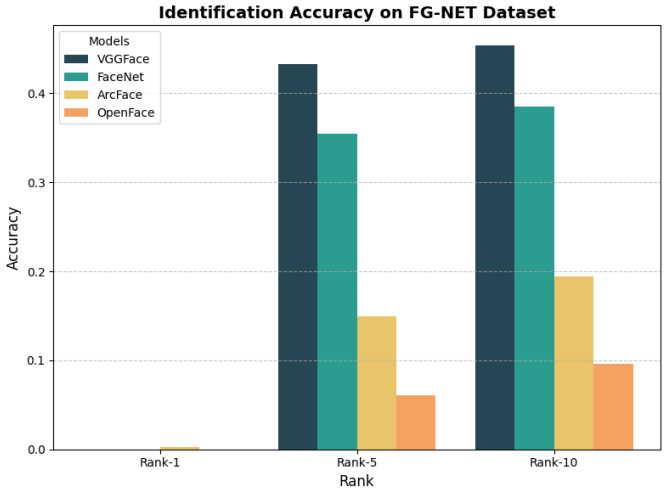

Fig. 1. Rank based identification accuracy on FG- NET dataset

VGGFace demonstrates highest identification accuracy across all ranks for FG-NET dataset. FaceNet performs competitively especially at higher ranks where its accuracy gets close to VGGFace. ArcFace shows moderate performance at higher ranks. OpenFace consistently produces lowest identification accuracy at all ranks.

For the CACD dataset, the trend remains similar. VGGFace outperforms all the models, with the accuracies increasing from 0.02 at Rank-1 to 0.42 at Rank-10. FaceNet shows a noticeable gap in comparison to VGGFace but still leads ArcFace, OpenFace. Both ArcFace and OpenFace show minimal accuracies across all ranks for CACD dataset.

In total, these results indicate that VGGFace is the most robust model, followed by FaceNet as a strong secondary performer. ArcFace and OpenFace still lag significantly, especially on the CACD dataset.

TABLE I
IDENTIFICATION ACCURACY ON FG-NET DATASET

Model	Rank - 1	Rank - 5	Rank - 10
VGGFace	0.0	0.433	0.453
FaceNet	0.0	0.354	0.385
ArcFace	0.002	0.149	0.194
OpenFace	0.0	0.060	0.096

TABLE II
IDENTIFICATION ACCURACY ON CACD DATASET

Model	Rank - 1	Rank - 5	Rank - 10
VGGFace	0.023	0.364	0.422
FaceNet	0.034	0.071	0.098
ArcFace	0.001	0.004	0.007
OpenFace	0.005	0.009	0.013

B. Verification Performance

The results for verification task has been summarized in terms of average AUC and EER value in Tables 3 and 4

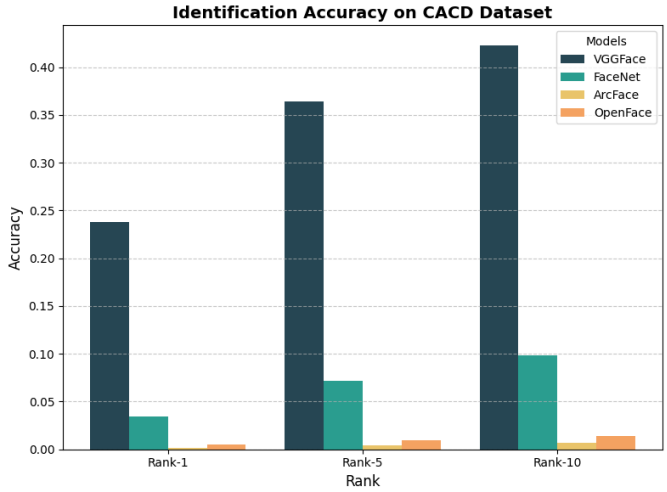

Fig. 2. Rank based identification accuracy on CACD dataset

respectively. Figures 3 and 4 show the AUC and EER heatmap for VGGFace which achieves the highest performance across all four models. Figures 5 and 6 indicate the AUC and EER heatmap for OpenFace model which demonstrates the weakest performance. VGGFace's effectiveness and OpenFace's suboptimal performance is consistent with the verification results. In both cases, the diagonal elements indicate the verification performance across the same age group, which consistently shows good performance. While non diagonal elements that show cross age verification face difficulty. All models, however, indicate a sharp drop in performance when tested with CACD dataset, likely due to the diverse nature of CACD, making it difficult for the models to generalize. It is also important to note that some age bin pairs lack verification due to lack of data, potentially impacting the accuracy of these results. These gaps highlight the need for balanced age distributions in future datasets. It also suggests that the model's improvements should be focused on generalization to more challenging and diverse populations.

TABLE III
AVERAGE AUC FOR EACH MODEL ON FGNET AND CACD DATASET

Model	FG-NET	CACD
VGGFace	0.813	0.523
FaceNet	0.787	0.519
ArcFace	0.664	0.505
OpenFace	0.590	0.503

TABLE IV
AVERAGE EER FOR EACH MODEL ON FGNET AND CACD DATASET

Model	FG-NET	CACD
VGGFace	0.212	0.486
FaceNet	0.226	0.487
ArcFace	0.348	0.495
OpenFace	0.419	0.499

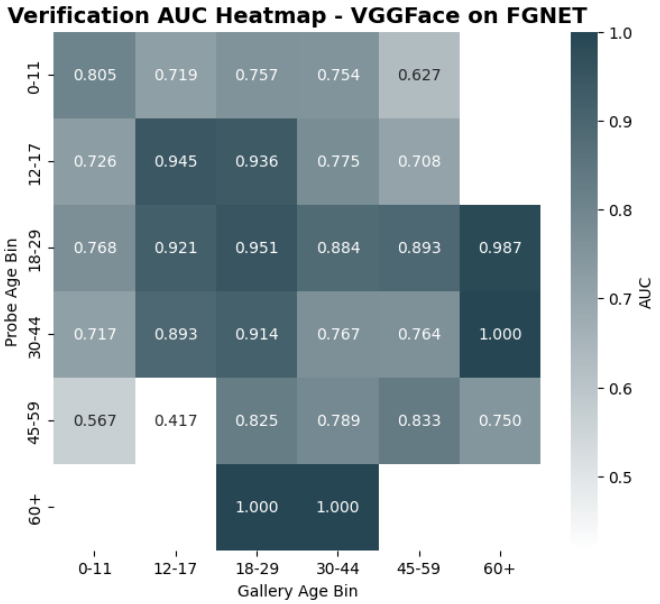

Fig. 3. Verification AUC Heatmap for VGGFace

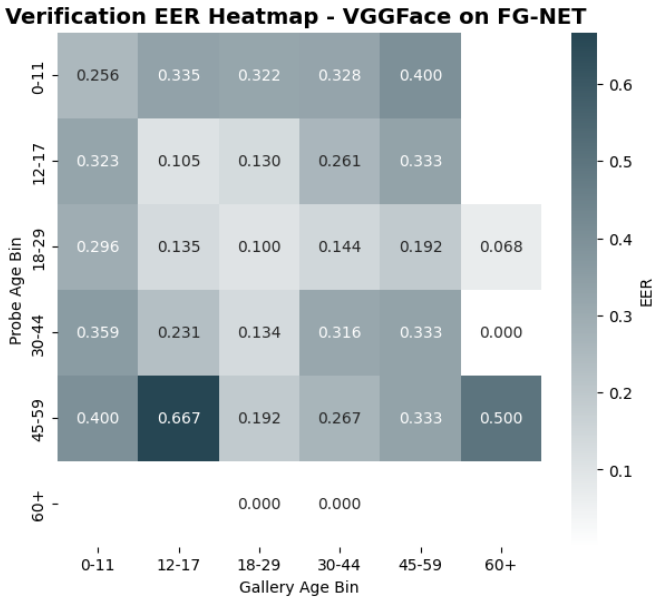

Fig. 4. Verification EER Heatmap for VGGFace

V. Discussion

A. Explanation for Low Rank-1 Accuracy

The Rank-1 metrics, as seen in table 1 and 2, are extremely low for both the datasets, and especially for the FG-NET dataset. This is due to the large age progression seen in FG-NET dataset. The embedding vector for a child will be very distant from the embedding vector for the same person as an adult. Another significant challenge is that a major portion of FG-NET dataset involves subjects transitioning from childhood through puberty, where rapid changes in facial

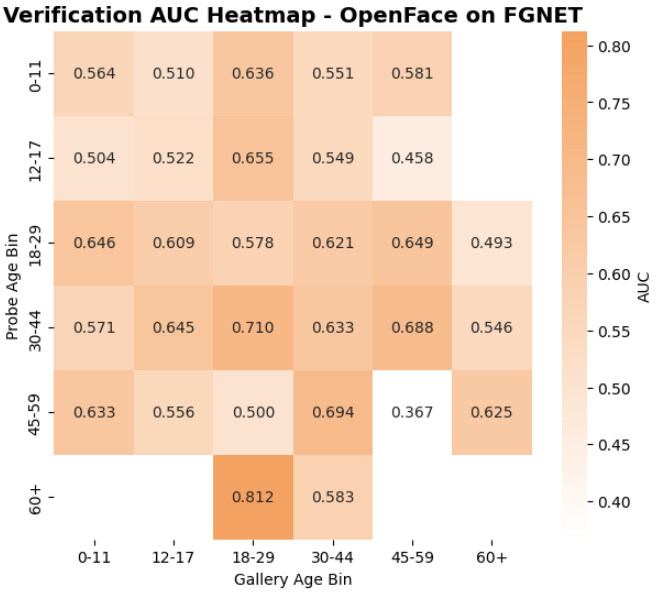

Fig. 5. Verification AUC Heatmap for OpenFace

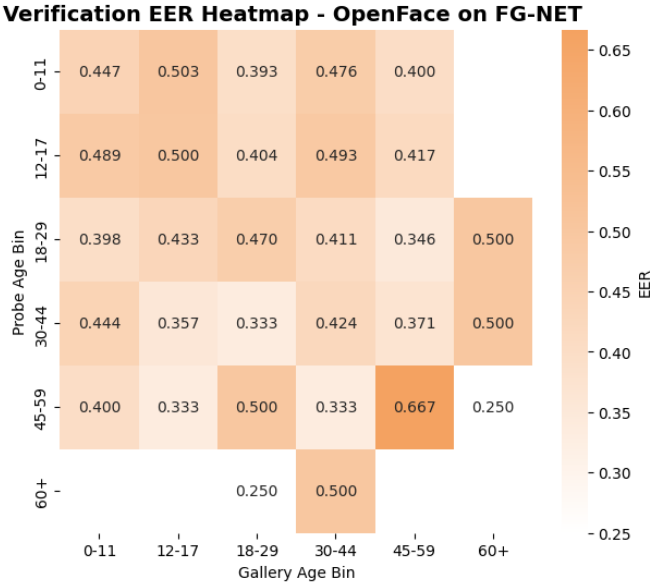

Fig. 6. Verification EER Heatmap for OpenFace

features tend to occur. However, these face recognition models are trained on massive datasets like MS-Celeb-1M which is heavily biased to adult faces. This confirms that baseline face recognition models are not inherently age invariant and need specific architectural or training modifications.

B. Model Wise Interpretations

The findings indicate that VGGFace outperforms the other evaluated face recognition models in this study. This can be attributed to the deep convolutional layers of the VGGFace model. The model utilizes a stack of 13 convolutional layers allowing it to capture minute facial features.

Additionally, its fully connected layers, each with a large number of channels help in capturing complex patterns. Moreover, the model is trained on a large dataset with over 2.6 million images comprising 2622 identities [13] which enhances the model's capabilities to generaliZe to new faces.

In contrast, OpenFace utilizes lightweight architecture developed for real time usage such as mobile applications. It uses fewer parameters and simpler network components making it computational efficient but less robust to detailed feature extraction. The model was also trained on a small dataset with approximately 500,000 images [14] which limits its ability to generalize.

The lightweight approach makes OpenFace suitable for applications that require real time face recognition and moderate accuracy rather than high precision face recognition systems that require deeper, and parameter rich models like VGGFace.

FaceNet incorporates two deep convolutional neural networks- Zeiler Fergus style network and an inception network. These help with feature extraction and residual connections.The use of triplet loss helps to cluster faces of the same person and separates those of different individuals in the embedding space, making verification highly efficient [15]. These characteristics make FaceNet ideal for real time verification tasks like access control.

ArcFace introduces an additive angular margin loss in its convolutional network training [16] , enhancing the discriminative power of face embeddings. This results in better separation between identities and improved performance with fewer parameters. Therefore, it is optimal in cases that need automated grouping and quick face searches like biometric authentication.

VI. Conclusion

Through this study we draw the following conclusions. First the transition from childhood to adulthood represents the most challenging scenario for all evaluated models. Second, VGGFace demonstrated the best performance, due to its deep convolution layers and large training dataset, making it well suited for high precision face recognition applications.

OpenFace's lightweight design prioritizes real time performance making it ideal for mobile or resource constrained environments.

Therefore, the overall choice of model depends on the specific application requirements such as need for real time processing, precision. Future work could extend this analysis to include a broader range of models and assess performance under challenging real-world conditions as discussed in section VII.

VII. Future Work

Future work could explore several directions to enhance the finding of this study. First, future work could include a broader set of face recognition models from libraries such as DeepFace or InsightFace which will allow for a more comprehensive comparison and validation of the results. Second, the current experiment for the identification task was conducted using

closed set protocols. Extending the evaluation to open set scenarios could better stimulate real world conditions where unseen identities could appear during testing. Third, performance could be assessed under realistic variations such as occlusions, lighting changes and masked faces to measure the robustness of these models.

Collectively, these initiatives could contribute to a more comprehensive understanding about the strengths and weaknesses of state-of-the-art recognition models and pave the way for development of more robust face recognition systems.

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